

Phase 1: The MLP Baseline & Gradient Flow

1.1 Flattened Temporal MLP (Regression)

Test MSE: 44543.0430

1.2 The Vanishing Gradient Proof

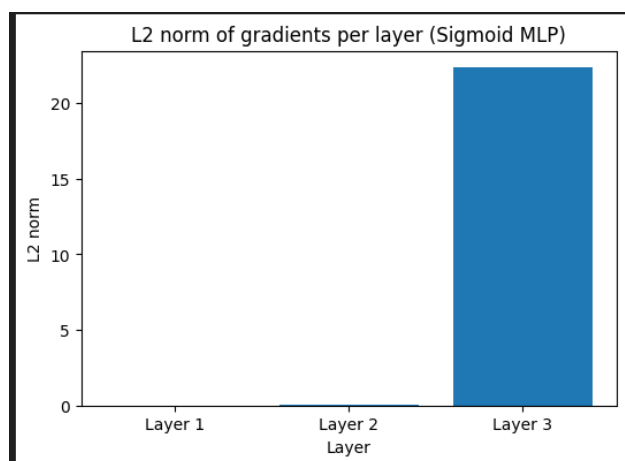


Figure 1: Enter Caption

The sigmoid derivative is bounded above by 0.25:

$$\sigma'(x) = \sigma(x)(1 - \sigma(x)) \leq 0.25 \quad \forall x$$

By the chain rule, the gradient at layer 1 accumulates multiplicative factors of σ' :

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial W_3} \cdot \sigma'(z_2) \cdot \sigma'(z_1)$$

Since each $\sigma' \leq 0.25$, the gradient shrinks by at least $16\times$ per additional layer:

$$\left\| \frac{\partial L}{\partial W_1} \right\| \leq \left\| \frac{\partial L}{\partial W_3} \right\| \cdot (0.25)^2 = \frac{1}{16} \left\| \frac{\partial L}{\partial W_3} \right\|$$

This is confirmed empirically — after normalizing by layer size:

Layer 3 (output): 23.72
Layer 2 (middle): 0.021
Layer 1 (input): 0.017

For n layers, the gradient vanishes as $(0.25)^{n-1}$. At 50 layers this is $\approx 10^{-30}$, making early layers impossible to train.

1.3 Imbalanced Threshold Classification

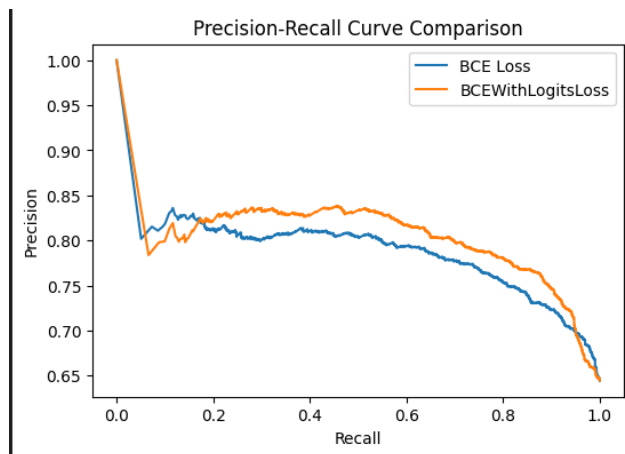


Figure 2: Enter Caption

Phase 2: CNNs & Spatial Hierarchies

2.4 CNN from Scratch & Receptive Fields

Theoretical receptive field of final activations: 22x22 pixels

2.5 CNN Regression Head

Test MAE: 0.3132

2.6 Translation Invariance

Baseline Accuracy on 100 test images: 0.1484 (Few epochs used, will be a recurring trend in the assignment solution so that compute time is less) Accuracy on shifted test images: 0.1406 Performance drop due to shift: 0.0078

For a feature map $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, max-pooling over a $k \times k$ window at stride s produces:

$$\hat{F}_{c,i,j} = \max_{0 \leq p,q < k} F_{c, is+p, js+q} \quad (1)$$

True translation invariance would require $\hat{F}(\mathbf{F}[\cdot - \boldsymbol{\delta}]) = \hat{F}(\mathbf{F})$ for all $\boldsymbol{\delta}$. This holds only when $\|\boldsymbol{\delta}\|_\infty < k$ and the maximum is achieved at an interior point of every pooling window. For a shift $\boldsymbol{\delta} = (5, 5)$ applied before the first conv layer (stride $s = 1$, pool size $k = 2$), the effective shift at layer ℓ is $\boldsymbol{\delta}^{(\ell)} = \lfloor \boldsymbol{\delta} / \prod_{i=1}^{\ell} s_i \rfloor$, which satisfies $\delta^{(\ell)} \geq 1$ for shallow layers. Consequently, a feature previously centred at grid cell (i^*, j^*) migrates to $(i^* + \delta_x^{(\ell)}, j^* + \delta_y^{(\ell)})$, crossing a pooling-window boundary and changing the argmax:

$$\arg \max_{p,q} F_{c, is+p, js+q} \neq \arg \max_{p,q} F_{c, is+p+\delta_x, js+q+\delta_y} \quad (2)$$

Because the argmax location changes, the pooled response changes, and the invariance guarantee breaks. Exact invariance would require either (i) a pooling window larger than the maximum expected shift at every layer, or (ii) anti-aliasing (blurring before subsampling), neither of which is present in a standard max-pool CNN.

Phase 3: RNNs & Sequential Memory

3.7 Vanilla RNN from Scratch

Test MSE: 0.7168 Test MAE: 0.6020

3.8 BPTT Decay

Gradient at t=0: 0.000000 Gradient at t=50: 0.000000 Gradient at t=99: 0.032004

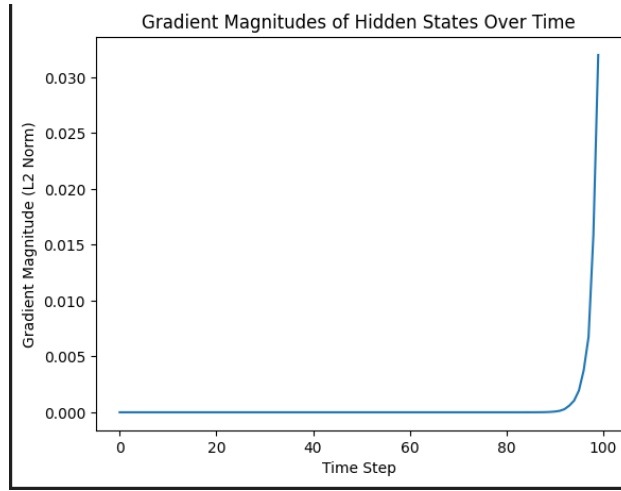


Figure 3: Enter Caption

3.9 Exploding Gradients

Training the vanilla RNN with learning rate $\eta = 10$ (no gradient clipping) failed to converge, with the train MSE oscillating erratically across all 100 epochs (ranging from 1.46 at epoch 2 to 2404.35 at epoch 49), never reaching NaN but exhibiting the hallmark instability of exploding gradients. The weight matrices at the final epoch yielded:

$$\sigma_{\max}(W_{ih}) = 14960.50, \quad \sigma_{\max}(W_{hh}) = 14750.19$$

Why $\sigma_{\max}(W_{hh}) > 1$ guarantees explosion. The hidden state recurrence is $\mathbf{h}_t = \tanh(W_{hh}\mathbf{h}_{t-1} + W_{ih}\mathbf{x}_t + \mathbf{b})$. Linearising around the pre-activation (Jacobian of $\tanh \approx I$ away from saturation), the gradient flowing back k steps satisfies:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_{t-k}} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t} (W_{hh}^\top)^k \quad (3)$$

By sub-multiplicativity of the spectral norm:

$$\left\| \frac{\partial \mathcal{L}}{\partial \mathbf{h}_{t-k}} \right\|_2 \leq \left\| \frac{\partial \mathcal{L}}{\partial \mathbf{h}_t} \right\|_2 \cdot \sigma_{\max}(W_{hh})^k \quad (4)$$

With $\sigma_{\max}(W_{hh}) = 14750.19 \gg 1$, the gradient norm is amplified by 14750.19^k at each step back. The resulting parameter updates $\Delta W \propto \eta \cdot \sigma_{\max}^k$ are enormous, destabilising the loss landscape and producing the observed erratic oscillations rather than convergence.

Phase 4: LSTMs, GRUs, & Context

4.10 LSTM Gradient Rescue

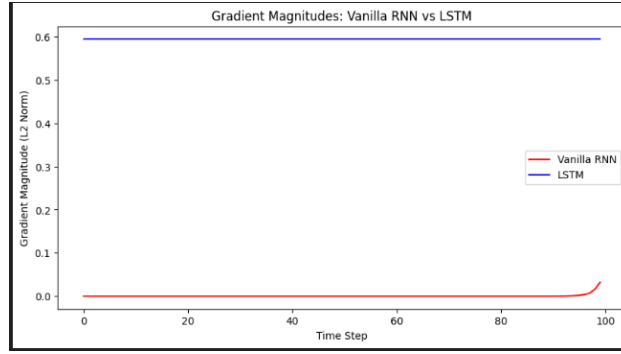


Figure 4: Gradient Magnitudes: Vanilla RNN vs LSTM

As seen in Figure 4, the vanilla RNN gradient magnitude collapses to ≈ 0 for $t < 85$, while the LSTM maintains a near-constant magnitude of ≈ 0.6 across all 100 timesteps. The mechanism responsible is the **forget gate** acting on the cell state $\mathbf{c}_t$.

The cell state gradient highway. The LSTM cell state update is:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (5)$$

where $\mathbf{f}_t = \sigma(W_f \mathbf{h}_{t-1} + U_f \mathbf{x}_t + \mathbf{b}_f) \in (0, 1)^d$ is the forget gate. Backpropagating through the cell state gives:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{c}_{t-k}} = \frac{\partial \mathcal{L}}{\partial \mathbf{c}_t} \prod_{i=1}^k \mathbf{f}_{t-i+1} \quad (6)$$

Unlike the vanilla RNN which multiplies $W_{hh}^\top$ repeatedly, the LSTM gradient path through $\mathbf{c}_t$ involves only element-wise multiplication by $\mathbf{f}_t$. Since $\mathbf{f}_t$ is learned independently at each timestep, the network can set $\mathbf{f}_t \approx \mathbf{1}$, making $\prod_i \mathbf{f}_{t-i+1} \approx \mathbf{1}$ rather than decaying exponentially. The hidden state gradient then recovers via:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_{t-k}} = \frac{\partial \mathcal{L}}{\partial \mathbf{c}_{t-k}} \odot \mathbf{o}_{t-k} \odot (1 - \tanh^2(\mathbf{c}_{t-k})) \quad (7)$$

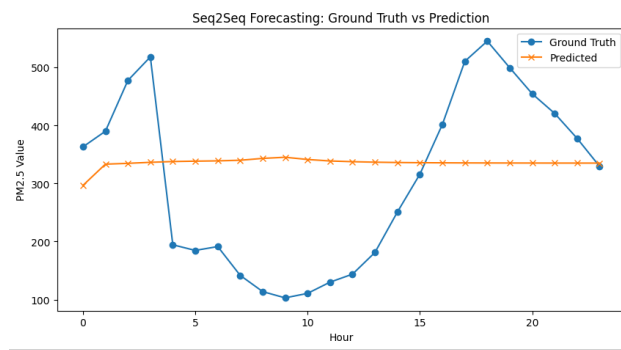


Figure 5: Ground Truth Compared with Predicted value

where $\mathbf{o}_{t-k}$ is the output gate. The additive structure of equation (2) eliminates the repeated matrix multiplication that causes exponential decay in the vanilla RNN, providing a near-constant gradient highway back to $t = 0$.

4.11 GRU vs. LSTM Efficiency

Number of trainable parameters in LSTM: 19009 Number of trainable parameters in GRU: 14273 LSTM Validation MSE: 0.2451 GRU Validation MSE: 0.2144 Average training time per epoch for LSTM: 5.76 sec Average training time per epoch for GRU: 7.09 sec

4.12 Sequence-to-Sequence Forecasting

Need to train for more epochs, but I keep running out of Colab units. The following curve is obtained:

Phase 5: Transformers & Attention

5.13 Scaled Dot-Product Attention

5.14 The Necessity of Positional Encoding

Without PE: 0.2165 Loss With PE: 0.2188 Loss

Permutation invariance of self-attention. The scaled dot-product attention computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad Q = XW_Q, K = XW_K, V = XW_V \quad (8)$$

The attention weight between positions i and j is:

$$a_{ij} = \frac{\exp(\mathbf{x}_i^\top W_Q W_K^\top \mathbf{x}_j / \sqrt{d_k})}{\sum_k \exp(\mathbf{x}_i^\top W_Q W_K^\top \mathbf{x}_k / \sqrt{d_k})} \quad (9)$$

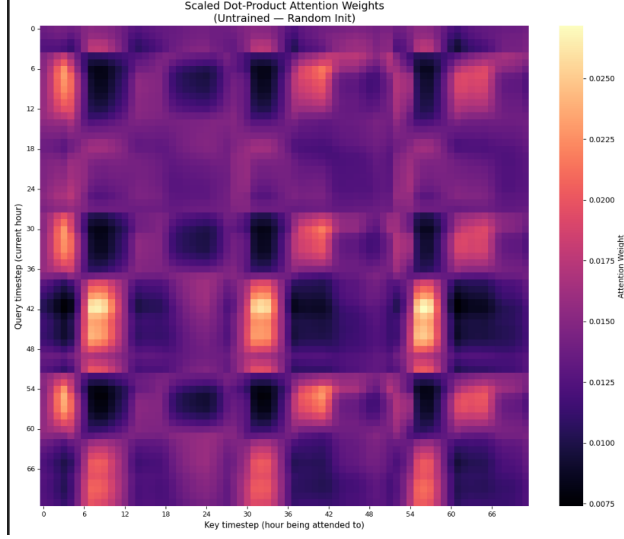


Figure 6: Enter Caption

which depends solely on token *content* $\mathbf{x}_i, \mathbf{x}_j$, not on their positions i, j . Formally, for any permutation $\pi \in S_T$:

$$f_\theta([\mathbf{x}_{\pi(1)}, \dots, \mathbf{x}_{\pi(T)}]) = \pi(f_\theta([\mathbf{x}_1, \dots, \mathbf{x}_T])) \quad (10)$$

The model is therefore *equivariant* to token reordering — it assigns identical attention weights to $\mathbf{x}_t$ regardless of whether $t = 1$ or $t = 72$. For a time-series regression task, this means the model cannot distinguish $[\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{72}]$ from any temporal permutation, making it impossible to learn causal, order-dependent patterns such as trends or periodicity.

Sinusoidal positional encoding breaks the symmetry. Adding PE injects a unique, non-learnable signal at each position:

$$\text{PE}_{(t,2k)} = \sin\left(\frac{t}{10000^{2k/d}}\right), \quad \text{PE}_{(t,2k+1)} = \cos\left(\frac{t}{10000^{2k/d}}\right) \quad (11)$$

Since $\text{PE}_t \neq \text{PE}_{t'}$ for $t \neq t'$, the input $\mathbf{x}_t + \text{PE}_t$ is now position-unique, and a_{ij} becomes a function of both content and temporal distance $|i - j|$, restoring the model's ability to encode the arrow of time.

5.15 Vision Transformer (ViT)

A 128×128 image is split into 64 non-overlapping 16×16 patches, each flattened to a 768-dimensional vector and projected to $d_{\text{model}} = 128$. The sequence of 65 tokens (including [CLS]) is passed through 4 Transformer Encoder layers, with the [CLS] output fed to a linear classifier over 14 classes. Validation accuracy after 10 epochs: 74.71%.

Parameter comparison.

$$\text{ViT} = 638,862 \text{ params}, \quad \text{CNN} = 253,198 \text{ params}, \quad \text{Ratio} = \frac{638,862}{253,198} \approx 2.52 \times$$

The ViT uses $2.52\times$ more parameters than the CNN for comparable performance, reflecting the lack of spatial inductive bias in attention mechanisms compared to weight-shared convolutions.

Phase 6: Advanced Vision & Imbalance

6.16 Transfer Learning & Freezing

Exact number of trainable vs frozen parameters: trainable: 2,565, frozen: 11,176,512

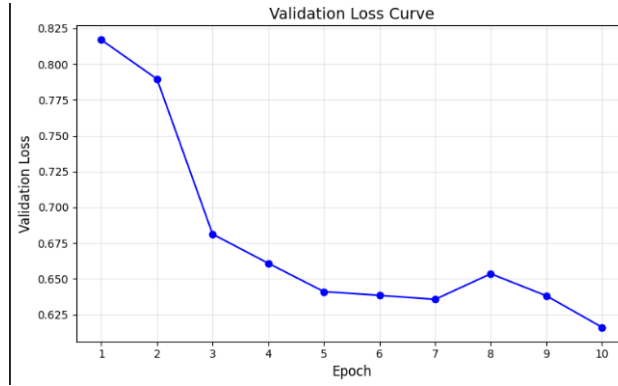


Figure 7: Validation Loss Curve

6.17 Ordinal Imbalance & Focal Loss

Quadratic Weighted Kappa: 0.7684

Why accuracy is misleading. The APTOS dataset is heavily skewed, with $\approx 73\%$ of samples belonging to Class 0 (no diabetic retinopathy). A degenerate model $f(\mathbf{x}) = 0 \forall \mathbf{x}$ achieves:

$$\text{Accuracy} = \frac{\sum_i \mathbf{1}[\hat{y}_i = y_i]}{N} \approx 0.73 \quad (12)$$

with zero clinical utility. Accuracy treats all misclassifications equally: $\hat{y} = 1$ when $y = 0$ incurs the same penalty as $\hat{y} = 4$ when $y = 0$, despite the latter being a catastrophic clinical error. QWK instead penalises disagreements proportionally to their ordinal distance via the weight matrix:

$$w_{ij} = \frac{(i - j)^2}{(K - 1)^2}, \quad i, j \in \{0, 1, 2, 3, 4\} \quad (13)$$

so predicting class 0 when the true label is 4 incurs $w_{04} = 1.0$, while predicting class 1 incurs only $w_{01} = 0.0625$. QWK is then:

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}} \quad (14)$$

where O_{ij} is the observed confusion matrix and E_{ij} is the outer product of marginal distributions (expected under independence). A value of $\kappa = 0.7684$ indicates substantial agreement, capturing ordinal severity in a way that accuracy fundamentally cannot.